# Xi'an Jiaotong-Liverpool University

# 西交利物浦大學

| PAPER CODE | EXAMINER | DEPARTMENT | TEL |
|---|---|---|---|
| INT202 | | | |

**2nd SEMESTER 2023/24 RESIT EXAMINATION**

**UNDERGRADUATE – YEAR 3**

**COMPLEXITY OF ALGORITHMS**

**TIME ALLOWED: 2 HOURS**

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**INSTRUCTIONS TO CANDIDATES**

1. **This is a closed-book examination.**
2. **The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
3. **Total marks available are 100. This exam will count for 100% of the final mark.**
4. **Answer all questions. There is NO penalty for providing a wrong answer.**
5. **Only English solutions are accepted.**

**Question 1. [8+12=20 MARKS]**

A) Consider the following code fragment.

```
for i=1 to n do
    for j=i to 2*i do
        output foobar
```

Let T(n) denote the number of times `foobar' is printed as a function of n.

1) Express T(n) as a summation (actually two nested summations). **[3 marks]**

2) Simplify the summation and give the worst-case running time using Big-O notation. **[5 marks]**

B) The processing time of a sorting algorithm is described by the following recurrence equation (c is a positive constant):

$T(n) = 3T(n/3) + 2cn;$
$T(1) = 0$

1) Solve this equation to derive an explicit formula for T(n) assuming n = 3m and specify the Big-O complexity of the algorithm. **[6 marks]**

2) Find out whether the above algorithm is faster or slower than usual Mergesort that splits recursively an array into only two subarrays and then merges the sorted subarrays in linear time cn. **[6 marks]**

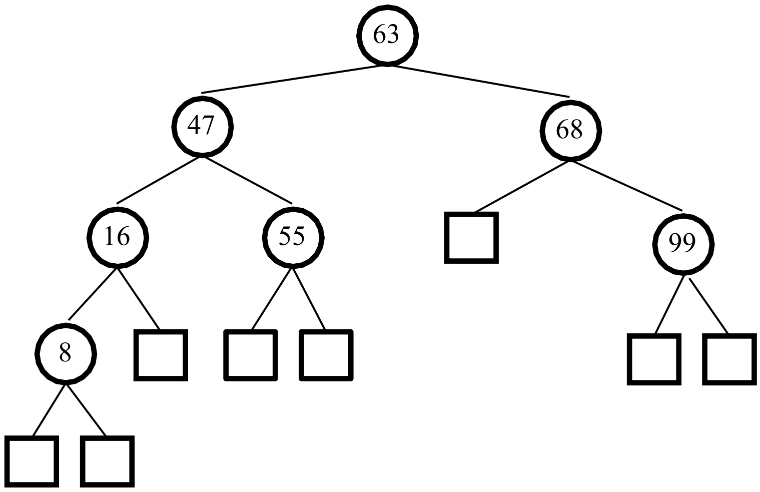
**Question 2. [5+10=15 MARKS]**

A) Consider a binary heap. Print the keys as encountered in a preorder travel. Is the output sorted? Justify your answer. **[5 marks]**

B) Convert an array [8, 19, 56, 88, 13, 5, 3, 33, 66, 40] into a maximum heap H. Draw the initial array and the percolation steps of creating the maximum heap. **[10 marks]**

**Question 3. [5+10=15 MARKS]**

Use the following graph for this problem.

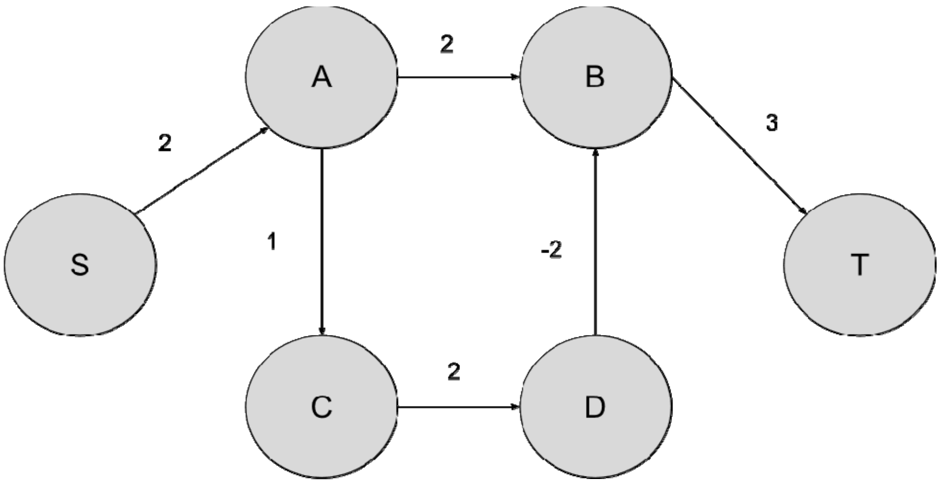



A) Draw the AVL tree resulting from the insertion of an item with a key 32 into the given binary tree. **[5 marks]**

B) For the AVL tree with key 32 inserted, draw the AVL tree after the elements 99 removed. Please show all intermediate steps. What is the running time of a removal operation in an AVL tree (using Big-O notation)?
**[10 marks]**

**Question 4. [10+5=15 MARKS]**

For a given weighted directed graph and two nodes s and t, the shortest path problem is to find a simple path (without repeating nodes) from S to T such that the sum of weighted edges in the path is smallest.




(1) Give a step by step process of using Bellman-Ford algorithm for finding a shortest path from S to T in the given graph. **[10 marks]**

(2) What is the shortest simple path and its' value? **[5 marks]**

**Question 5. [5+5+10=20 MARKS]**

(1) Find the multiplicative inverse of 5 mod 91. **[5 marks]**

Alice and Bob wish to use the RSA cryptosystem for communication. Alice announced the public modulus and encryption key pair to be (91, 5).

(2) Assume Bob wants to send a message 4, what's the encrypted message? **[5 marks]**

(3) What is the private key? **[10 marks]**

**Question 6. [5+10=15 MARKS]**

A *Clique* C of an undirected graph G = (V, E) is a subgraph of G such that all pairs of two vertices in C are adjacent to each other. In other words, it is a complete subgraph.
The Clique decision problem asks "Given an undirected graph G = (V,E) and a positive integer k, and the problem is to check whether a clique of size k exists in G."

(1) Prove that the Clique decision problem is in NP. **[5 marks]**

(2) Provide a reduction from the Vertex Cover decision problem to the Clique decision problem, and prove its' correctness. **[10 marks]**

**END OF EXAM PAPER**